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EFFECT OF REPETITIVE INFANT–INFANT SEPARATION OF YOUNG MONKEYS[1]

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Virtually all previous studies investigating the effects of separation from an attachment object have used a mother–infant model. In view of several methodological difficulties inherent in mother–infant separation, an alternative procedure, that of repetitive, short-term, infant–infant separations, is proposed. Results from three studies employing this procedure with young monkeys indicated that: (*a*) each short-term infant–infant separation produced behavior patterns similar to those exhibited by infants separated from their mothers; (*b*) *S*s did not adapt to the separation procedure, but rather continued to show similar reactions to each successive separation; and (*c*) a cumulative effect of repetitive infant–infant separations was an arrest of maturation of social development in the monkey *S*s. Implications of the findings are discussed.

Beyond question, the social relationship that has received the most attention from psychologists is the mother–infant dyad. Justification for this emphasis comes from ethological, philosophical, and methodological considerations. It is generally accepted that among all mammalian species, an infant's first social responses are directed toward its mother or mother-substitute. Hence, for any given infant, the relationship with the maternal object is assumed to be first chronologically, highest in probability of occurrence, and freer from effects of experience than any possible subsequent social relationship it may enter during its lifetime. The process by which an infant becomes attached to its mother has been investigated to generate normative developmental data (e.g., Shaffer & Emerson, 1964), to develop general theories of attachment (e.g., Bowlby, 1958), to offer hypotheses concerning the way the young of any species become a socialized part of their living group (e.g., Jones & Gerard, 1967), and to serve as a general model for the formation of social relationships later in life (e.g., Gewirtz, 1961). Studies of these types focus on the normal development of the infant–mother affectional bond.

There have also been a number of investigations of occasions in which this normal pattern has been interrupted—more specifically, where the infant has been separated from its mother. In the human literature, these consist primarily of observational studies of hospitalized children or orphans. Among the most spectacular of these studies are the reports of Spitz (1946) and Bowlby (1960) of behavior syndromes exhibited by children separated from their mothers early in life. Although the children observed by Spitz were between 6 mo. and 1 yr. of age while those observed by Bowlby were considerably older, between 2 and 5 yr. of age, the descriptions of the *S*s' behavior during the period of separation from the mother were qualitatively similar. Both reported an initial period of active protest or agitation, characterized by marked increases in such vocalizations as crying and screaming and in activity. This behavior did not persist, but instead was followed by a stage which Spitz called "withdrawal" and Bowlby termed "despair," embodying dejection, stupor, retardation of development, retardation of reaction to stimuli, marked decrease in activity, and withdrawal from the environment. It was these behaviors, along with "the physiognomic expression which . . . would in an adult be described as depression [Spitz, 1946, p. 316]," which led Spitz to term the general syndrome "anaclitic depres-

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sion." Spitz reported almost complete and immediate recovery in his children upon return to their mothers, whereas Bowlby observed a third stage which he called "detachment," involving hostility toward the mother upon reunion. This inconsistency may have resulted from the difference in age of the children observed by the two authors.

Systematic study of human mother–infant separation is handicapped by a number of virtually uncontrollable factors. An investigator's *S* population is determined by what society provides rather than by his experimental needs. As a result, strict control of variables such as age of *S*s, onset and duration of separation, and *S* environment prior to, during, and after separation is often impossible. In the case of hospitalized children, the physical condition of the infant requiring hospitalization undoubtedly confounds pure effects of separation. Moreover, what observations an investigator makes and how he records them are usually and properly determined by the attitudes of parents and hospital or orphanage personnel rather than by his own research needs or desires. In short, ethical considerations virtually prohibit experimental manipulation and severely constrain observational study of human mother–infant separation. It is not surprising that investigators have turned to other species to investigate the phenomenon.

During the past 10 yr., a number of studies employing nonhuman primates as *S*s have been carried out with the sole intention of assessing the effects of mother–infant separation (e.g., Hinde, Spencer-Booth, & Bruce, 1966; Jensen & Toleman, 1962; Kaufman & Rosenblum, 1967; S. Peters & J. Rosevear, unpublished data, 1969; Seay, Hansen, & Harlow, 1962; Seay & Harlow, 1965). The general paradigm of all these efforts has been similar: rear the infant with its mother for a specified period of time, separate for a specified period of time, then reunite the pair, carefully observing behaviors prior to and during separation and following reunion. The studies differ in respect to species employed (rhesus versus pigtail macaques), age at separation (4.5–11 mo.), type of separation (total social versus partial social, denying physical but not visual or auditory contacts), duration of separation (5 min. to 6 wk.), number of separations (1–3), and social environment during separation (alone versus infant group versus colony with mother removed). Yet the findings are amazingly consistent in the qualitative sense. Upon separation, infant monkeys always exhibit an initial period of extreme distress and agitation characterized by marked increases in vocalization and activity. This behavior seldom lasts more than 48 hr. and, if the separation is continued, is almost always followed by a period of striking reduction in vocalization, locomotor, exploratory, and play activity, and in some cases, by autistic posturing. Some investigators have called this behavior depression. Upon reunion, all investigators report a temporary increase in mother-directed behavior which eventually drops to or below preseparation levels. Bowlby's detachment phase has not been observed in any monkey *S*s. Although quantitative differences in *S* behaviors exist between studies, undoubtedly reflecting the variations in experimental procedure, the general effects of separation are similar enough between studies and to the observations of human infants separated from their mothers to justify use of nonhuman rather than human *S*s in separation studies to gain greater control of experimental variables.

Unfortunately, there are a number of factors inherent in the aforementioned experimental design which are less than desirable. The first involves the very act of physically separating a monkey infant from its mother. It is, at the very least, a most traumatic and exhausting procedure for both *S*s and *E*s. Generally, the mother, clutching the infant, is caught in a net and while two or more handlers restrain her, a third wrestles the infant from her grasp. Although the contribution of this act itself to the subsequent agitation exhibited by both mother and infant could probably be determined by catching mothers without removing their infants, the fact remains that it is a touchy procedure, involving considerable time, effort, and personnel.

A second factor concerns the behavior of the monkey mother. If one is interested in

the effect of separation upon the infant, it would be desirable to have the infant return to an attachment object similar to the one from which he was separated, or at least one who was similarly affected by the separation experience. Monkey mothers are in neither case ideal. As their infants mature, monkey mothers exhibit an increasing amount of active rejection and passive disinterest in their offspring (e.g., Harlow & Harlow, 1965). It is thus difficult to assess the true effect of prolonged separation on mother–infant behaviors following reunion, for the mother's behaviors are affected by two factors: the effect of this naturally occurring pattern of increasing rejection of the infant as well as the effect of separation per se. In addition, Jensen (1968) has presented evidence that monkey mothers are less adversely affected by the separation experience than are monkey infants.

A third disadvantage of single, long-term separation studies is that while the effects of separation are profound, they are not prolonged. The stage of protest or agitation, for example, rarely persists beyond 48 hr. Furthermore, Kaufman and Rosenblum (1967) reported large quantitative individual differences among *S* reactions to maternal separation. Hence, in order to accurately assess the continuous, quantitative effects of mother–infant separation, it is necessary to employ a large number of *S*s and to observe them almost constantly during the separation phase of the experiment in order to generate meaningful data.

In order to combat these problems, we propose two modifications of the general separation study design. First, rather than separating mothers from infants, we suggest separating infants, reared together without mothers from early in life, from each other. There is abundant evidence that monkeys raised only with peers develop strong attachments to each other and that infantile attachment behaviors such as clinging persist far longer than in monkeys reared with their mothers (e.g., Chamove, 1966; Harlow, 1961; Pratt, 1969). Monkey infants so reared do not actively reject their partners with age as do monkey mothers, but instead encourage additional attachment behaviors by reciprocation. Differences in reaction to separation among infant-only groups can be attributed to individual differences in *S*s rather than to a combination of individual and maturational differences as is the case in mother–infant groups. Finally, the physical act of separation of infants is quick and easy. It takes one *E* only a few seconds to separate an infant–infant pair.

A second modification involves the use of a large number of short-term separations rather than a single or few long-term separations. If *S*s do not adapt to the separation experience, each phase of the reaction to separation should be exhibited during the course of each separation. Thus, over a large number of separations, the total amount of time *S*s exhibit each stage of reaction to separation will be greater than for a single long-term separation, allowing an investigator to gather roughly equivalent data with fewer *S*s and less concentrated periods of observation.

The technique of repetitive short-term separation of infant-only groups was employed in three studies. One study, to be subsequently described in detail, involved separation of a group of four infants a total of 20 times, each separation lasting 4 days. Two minor studies will be cited briefly in the discussion to offer support for the findings of the major study.

METHOD

The *S*s were four rhesus monkeys, two males and two females, born within 4 days of each other. They were separated from their mothers at birth and were raised in a nursery for the first 15 days according to the methods described by Blomquist and Harlow (1961). Subsequently, they were placed in a large living cage, $6 \times 3 \times 4$ ft., containing two blocks of wood and two plastic balls, which served as play objects, and four cloth towels. The *S*s remained undisturbed in this cage as a group of four until their mean age was 90 days, at which point the first separation series was begun. This series consisted of 12 week-long separation cycles, in which *S*s were physically separated and individually housed in $2.5 \times 2 \times 2.5$ ft. cages for the first 4 days and then returned to the group cage for the remaining 3 days of each week. During the periods of separation, *S*s could see and hear each other, and each of their individual cages contained one of the play objects and one of the cloth towels. At the end of

the first separation series, *S*s, approximately 6 mo. old, were housed as a group of four in the large living cage for a period of 6 wk. A second separation series, identical in procedure to the first, was then begun, lasting a total of 8 wk. Following this second series, *S*s were housed in the large cage for an additional month. In all, *S*s were separated 20 times, each separation lasting 4 days, and were housed together the remainder of their first 10 mo. of life.

Systematic observation of the *S*s began 2 wk. prior to the start of the first separation series and continued throughout the experiment. Observational sessions lasted 30 min., during which time each *S* was watched and its behavior scored for 7.5 min. The measures taken consisted of frequency and duration during each 7.5-min. period of *S* behaviors falling into nine categories: ventral cling, self-clasp, self-mouth, rock and huddle, passive, locomotion, environmental exploration, social contact, and vocalization. A tenth category, sex and play, was added following the first separation series. A description of behavior encompassed in each category follows:

Ventral cling: contact of own ventral body surface with another *S*.
Self-clasp: clasping of any part of own body with hand(s) and/or foot (feet).
Self-mouth: oral contact to own body.
Rock and huddle: self-clasping accompanied by stereotypic rocking.
Passive: absence of all social, exploratory, and locomotor activity.
Locomotion: any self-induced change in physical location of self.
Environmental exploration: tactual and/or oral manipulation by *S* of inanimate objects, for example, towel, toy, cage lock, etc.
Social contact: tactual and/or oral contact with another *S* other than ventral clinging.
Vocalization: any sound emitted by *S*.
Sex and play: any sexual advances or posturing directed toward another *S* and/or any type of socially directed play activity (including rough-and-tumble approach–withdraw, and noncontact type play).

Two *O*s were employed, each scoring approximately half of the sessions. During the 2-wk. period prior to the start of the first separation series, there was 1 observational session per day. During the first separation series, there were 12 observational sessions per week, distributed as follows: 4 sessions during the first 24 hr. of separation, 4 sessions during the final 48 hr. of separation, and 4 sessions during the 72-hr. period that *S*s were together. For the remainder of the experiment, there was 1 observational session per day.

RESULTS

Three clear-cut findings emerged from analysis of the data. First, behavioral manifestations of the stages of reaction to separation from the mother described for both human and monkey infants were clearly exhibited during the weekly separation cycles by these *S*s. Second, *S*s did not adapt to the multiple separations. For each separation series, *S*s' reactions to the latter separations were at least as severe as their reactions to the first separations in the series. Finally, an overall effect of the multiple separations was an almost complete arrest of maturation of social behaviors of these *S*s. Their social behavior at age 9 mo., following the separations, was virtually identical to their behavior at age 3 mo., prior to the separations, in sharp contrast to the rapid and broad behavioral maturation traditionally reported for similar-age monkeys not subjected to separations.

Data accumulated from all *S*s during the first series of 12 separations and returns were divided into three groups, the first coming from observations taken during the first 24 hr. following each separation, the second from observations during the final 48 hr. of each weekly separation, and the third from observations during the 72 hr. each week that the animals were together. The mean durations per session of these three sets of observations are plotted category by category in Figure 1. It can be seen from the graph that the three behavioral profiles differ radically from one another, and this observation is supported statistically by two-tailed t tests of the category means. During the first 24 hr. of separation, *S*s exhibited significantly more locomotion, vocalization, and tactual and oral manipulation of the towels, play objects, and cage walls than during either the last 48 hr. of separation or the 72 hr. together (all $ps < .005$). During the last 48 hr. of separation, *S*s exhibited significantly more rock and huddling, self-clasping, and passive behavior than during the first 24 hr. of separation or the 72 hr. together (all $ps < .005$). During the weekly 72-hr. periods together, *S*s exhibited significantly less behaviors in all categories measured than during the other two periods (all $ps < .005$) with two exceptions, self-mouthing and environmental exploration. While together, *S*s self-mouthed more ($p < .005$) and explored the environment less ($p < .005$) than during the first 24 hr. of separa-

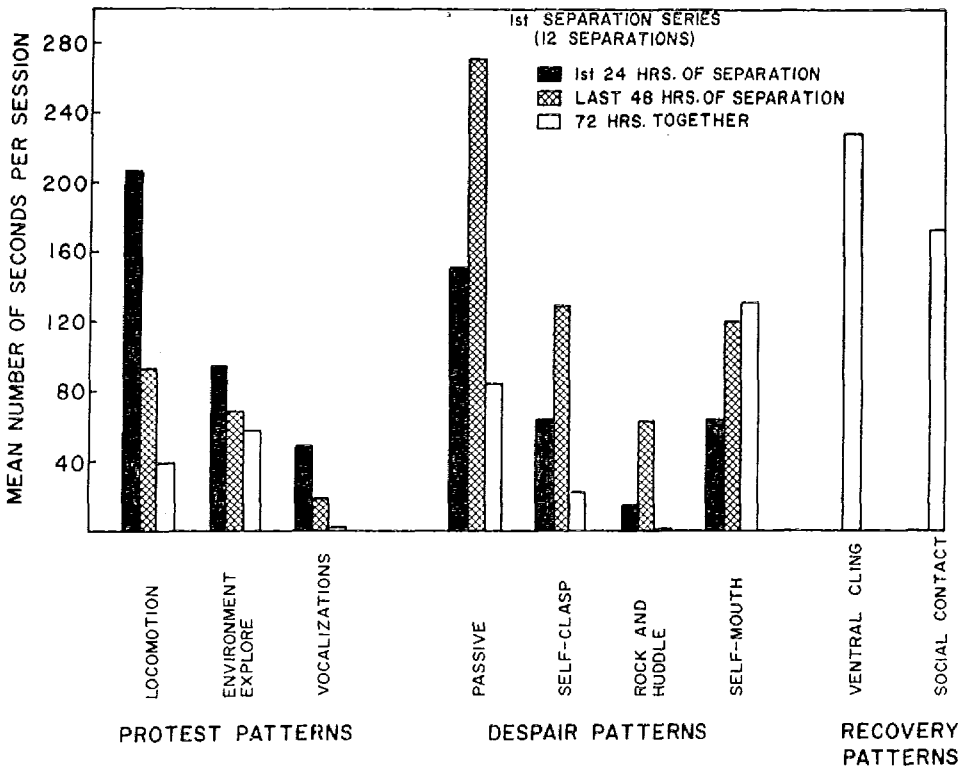


FIG. 1. Category means for the first separation series.

tion, but not during the last 48 hr. of separation. Most of the *Ss*' behavior while together fell into the two social categories, ventral cling and social contact. Thus, *Ss*' behavior was characterized by high levels of locomotion, environmental exploration, and vocalization during the first 24 hr. of the separations; by high levels of rock and huddling, self-clasping, and passivity during the latter part of the separations; and by high levels of ventral clinging and social contact while together, precisely those behaviors attributed in the mother–infant separation literature to the protest, despair, and recovery stages, respectively.

Figure 2 presents equivalent category means for the second separation series. Although the differences between stages do not appear to be as distinct as those in the first series, all significant relationships found in the first series were also found to be statistically significant in the second series, with three exceptions. First, no rock and huddling behavior was observed in any *Ss* during the second series. Also, the differences in environmental exploration and vocalization between the first 24 hr. and the last 48 hr. of the separations found to be significant in the first series fell short of acceptable levels of significance in the second separation series.

A second finding was that within each separation series, *Ss* did not adapt to the separation procedures. Their reactions to the latter separations of each series were at least as severe as their reactions to the first separations of each series. Figure 3 presents category means of the *Ss*' behaviors during the first six separations versus those in the last six separations for the first series. Two-tailed t tests of these means yielded no significant differences in vocalization, passive, and self-clasping behaviors between the first six and last six separation periods. Significant decreases in self-mouthing ($p < .05$), environmental exploration ($p < .01$), and locomotion ($p < .005$) were found, while a significant

increase ($p < .005$) occurred in rock and huddling. The decreases in environmental exploration and locomotion and the increase in rock and huddling suggest that *Ss* were showing less protest (although vocalization levels did not differ) and more despair-like behavior during the latter separations. An equivalent analysis for the second separation series, comparing behavior levels recorded during the first four separations of the series with those from the last four, yielded no significant differences for any behaviors observed (Figure 4).

The third and perhaps most interesting finding was the virtual arrest of normal maturation produced by the multiple separations. Figure 5 gives a category by category breakdown of the *Ss*' behavior at 3 mo. (prior to separation), at 6 mo. (following the first separation series), and at 9 mo. (following the second separation series). Two-tailed *t* tests indicated no significant differences in 3-, 6-, and 9-mo. levels of ventral cling and self-mouth, both neonatal behavior patterns, whereas both the 6- and 9-mo. levels of passivity, another neonatal pattern, were significantly higher than the 3-mo. level ($p < .025$ and $p < .005$, respectively). There were no significant differences in the three age levels of social contact, a more advanced neonatal pattern, while the 9-mo. level of locomotion was significantly lower than the 3-mo. level ($p < .025$), a reversal for the normal developmental pattern for young monkeys. Both the 6- and 9-mo. levels of self-clasping exceeded the preseparation level (both ps $< .05$). Finally, the advanced infantile patterns of environmental exploration and sex and play did not increase with age in these *Ss*. Rather, the 3-mo. level of environmental exploration exceeded the level at 9 mo. ($p < .05$), while there were no differences in the levels of sex and play. All in all, these data strongly suggest that *Ss* were not exhibiting the maturation of behavior patterns characteristic of monkeys their age. Neonatal behaviors

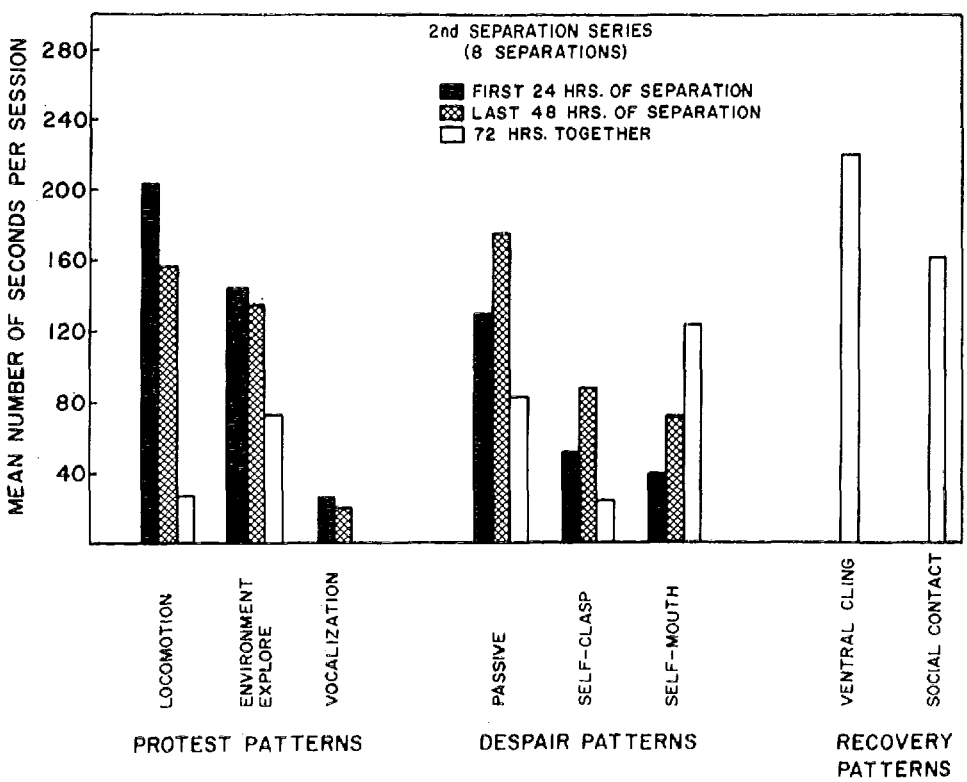


FIG. 2. Category means for the second separation series.

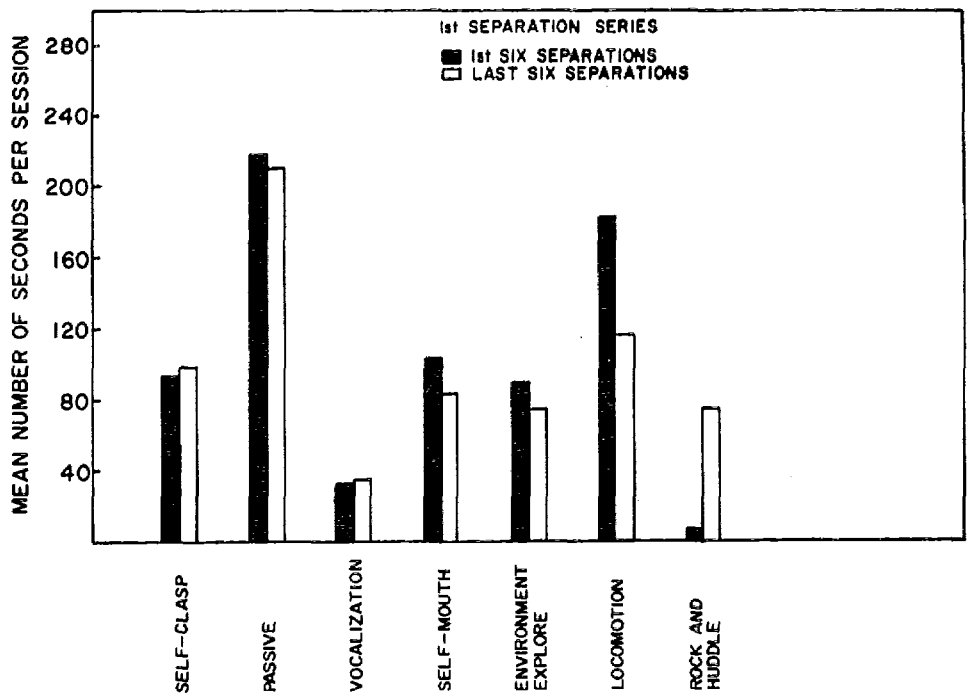


FIG. 3. Category means during first six versus final six separations in first separation series.

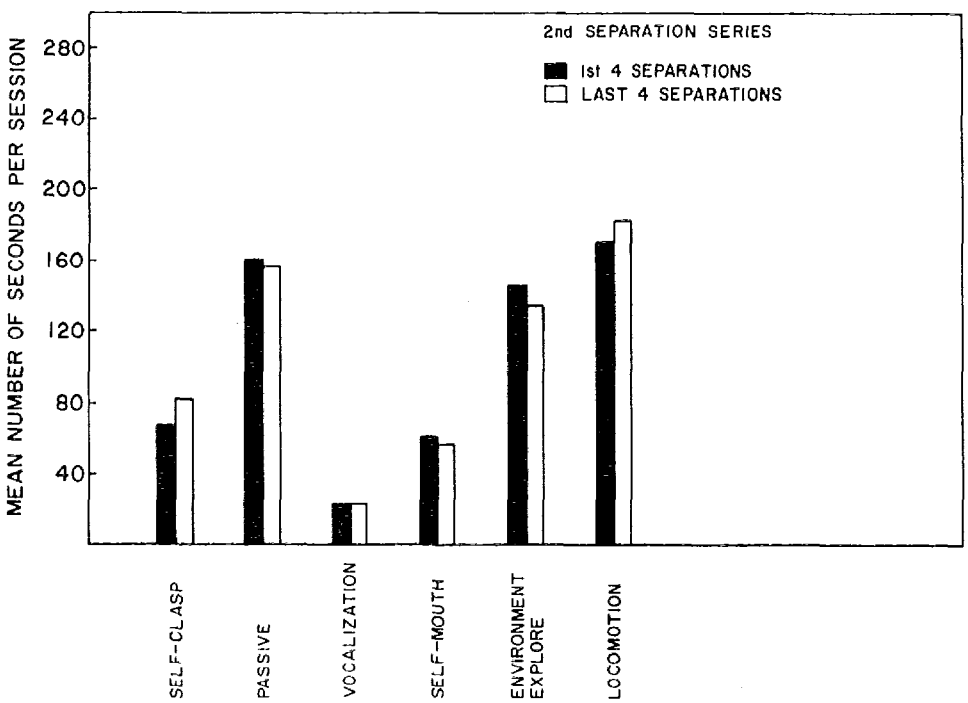


FIG. 4. Category means during first four versus final four separations in second separation series.

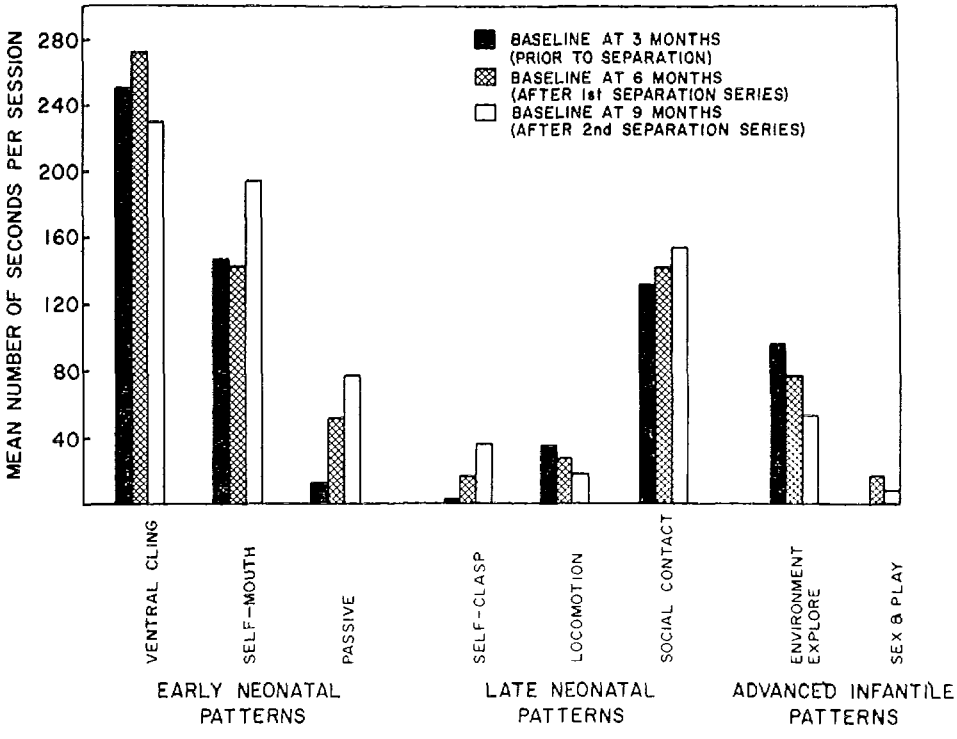


FIG. 5. Category means for 3-, 6-, and 9-mo. base-line observation periods.

did not decrease or disappear with age, but instead maintained their 3-mo. levels or increased. Behaviors characteristic of older infants did not increase with age for these *Ss*, but instead decreased or stayed near the preseparation levels.

DISCUSSION

The results of the preceding experiment give strong support to the notion that repetitive infant–infant separation is a legitimate technique not only for the study of reaction to separation from an attachment object, but also for other purposes such as arrest of social maturation. Unfortunately, the lack of both a second experimental and an appropriate control group of *Ss* warrants restraint in conclusions derived from these data alone. For this reason, we offer additional data accumulated from brief studies of two other groups of four young rhesus monkeys as supportive evidence for the above findings.

The first group of *Ss* was reared from birth in a large 4 × 4 × 6 ft. pen as a group of four mother–infant pairs. When their mean age was 8 mo., the infants were removed from their mothers and placed together in a 6 × 3 × 4 ft. cage where they remained undisturbed until their mean age was 13 mo. At this time they were subjected to a series of six separations and returns according to the previously described procedure, that is, they were separated and housed individually, maintaining visual and auditory contact with each other for 4 days and then returned to their group cage for 3 days. This cycle was repeated for 6 successive wk. The *Ss*' behaviors were scored on a daily basis in an identical manner as in the previous study, beginning 2 wk. before the first separation and ending 2 wk. following the final return. Figure 6 presents category means for the first 24 hr. of the six separations, the final 48 hr. of the six separations, and the 72 hr. *Ss* were together each week. Although less well differentiated, behavior patterns consistent with those patterns exhibited by *Ss* in the previous study were clearly found. The high-

est levels of locomotion, environmental exploration, and vocalization occurred during the first 24 hr. of the separations, the highest levels of passivity and self-clasp occurred during the latter 48 hr. of the separations, and the time together was characterized by a high level of ventral cling and relatively low levels of all other behaviors recorded. In addition, *Ss* showed a slight deepening of the despair phase and a lessening of the protest phase during the course of the study, as indicated by gradual increases in levels of passivity, self-clasp, and self-mouth and by gradual decreases in levels of environmental exploration and vocalization. This same phenomenon was observed during the course of the first separation series in the previous study.

As a whole, *Ss*' reactions to multiple, short-term infant–infant separations were remarkably similar to the reactions exhibited by the monkeys in the previous study, despite the fact that these *Ss* were considerably older before separation (13 mo. versus 3 mo.) and had been reared with their mothers for the first 8 mo. of life.

A second group of four infants was also reared from birth with their mothers in a 4 × 4 × 6 ft. pen. When the infants reached a mean age of 6 mo., they were removed from their mothers and placed in a 6 × 3 × 4 ft. cage as a group of four for 1 wk. They were then separated and housed individually in 2.5 × 2 × 2.5 ft. cages, permitting visual and auditory contact with each other, for 4 days, and then returned to the group cage for 1 wk., after which they were placed back in the pen with their mothers. Daily observations were recorded in the standard manner during the entire time *Ss* were separated from their mothers. Category means for the week following maternal separation (preceding infant–infant separation), for the first 24 hr. of infant–infant separation, for the last 48 hr. of infant–infant separation, and for the week together prior to return to the mothers are presented in Figure 7. Contrary to expectations, these *Ss* did not exhibit the typically

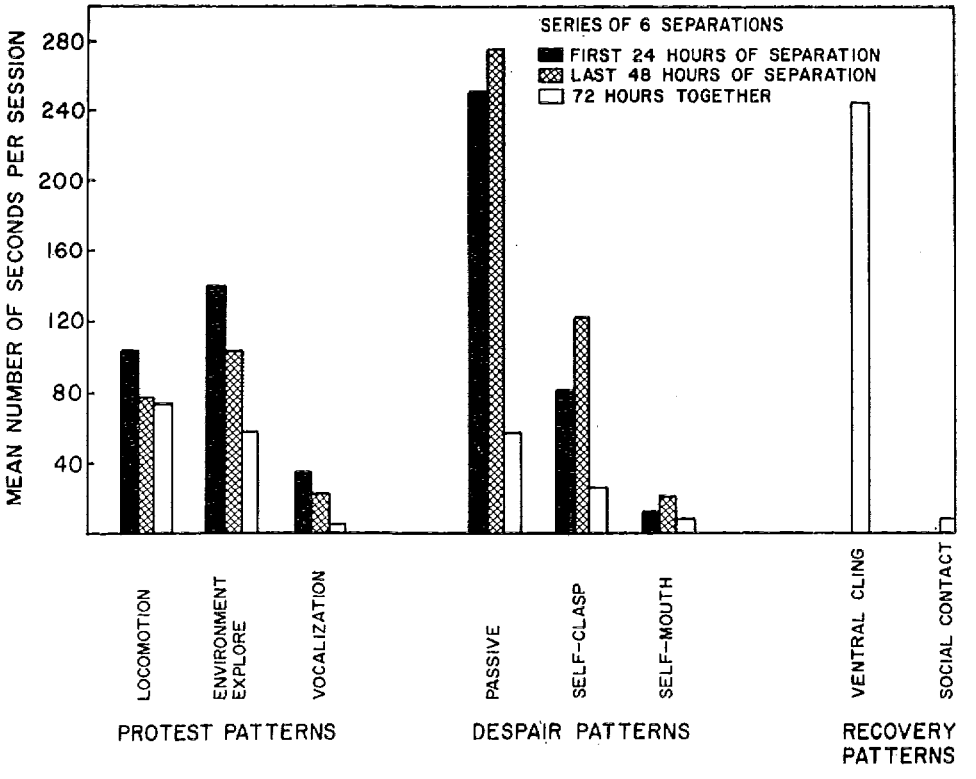


FIG. 6. Category means for series of six separations.

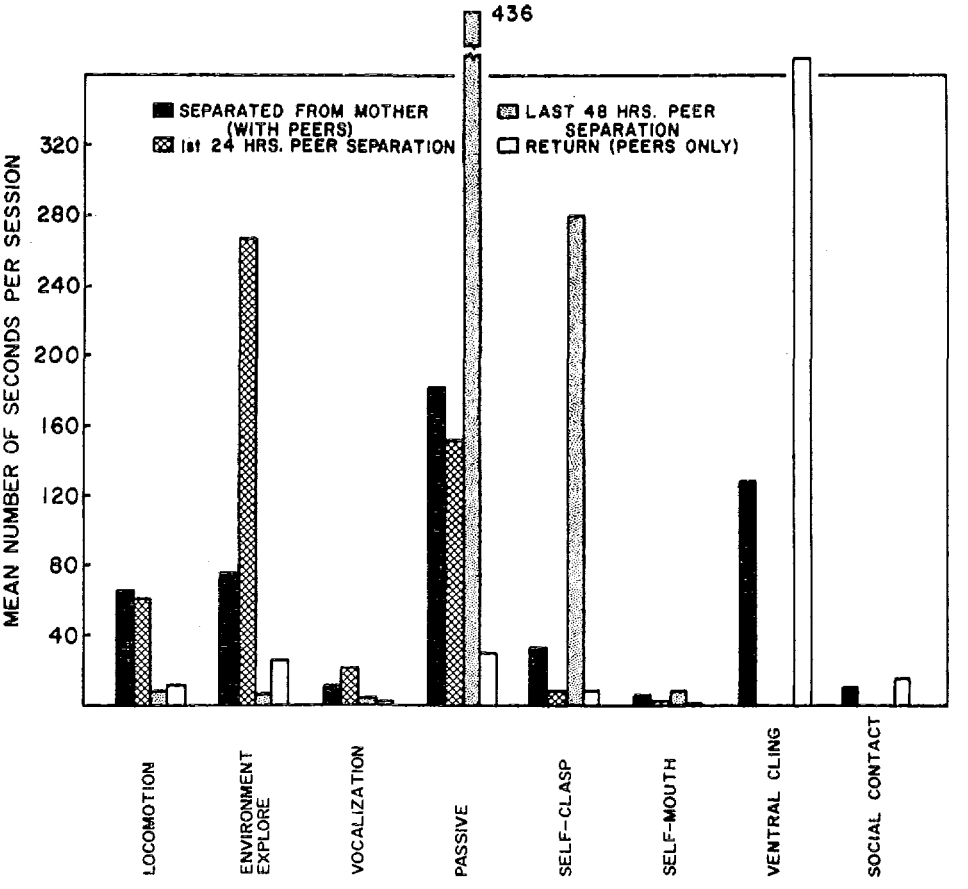


FIG. 7. Category means for *S* behaviors while separated from their mothers.

reported, violent reaction to maternal separation. Rather, they showed moderate levels of most behaviors scores, including a fairly substantial degree of ventral cling to each other. On the other hand, *S*s' reactions to infant–infant separation were anything but moderate. During the first 24 hr., they exhibited an unusual amount of environmental exploration and considerable vocalization. Even more striking was their observed behavior during the final 48 hr. of infant–infant separation. The *S*s spent less than 2% of the observation periods engaging in *any* behaviors other than passive or self-clasp. We can scarcely conceive of a more profound behavioral exhibition of so-called depression in monkeys. Finally, upon peer reunion, *S*s exhibited a striking rise in ventral cling over the preseparation level. While we hesitate to make any generalizations regarding the relative severity of reaction to infant–infant separation as opposed to mother–infant separation, we do feel that these *S*s definitely showed the protest-despair-recovery reaction to infant–infant separation over and above any effects produced by the prior separation from their mothers.

The data from these three studies give unequivocal support to the hypothesis that infant–infant separation of young rhesus monkeys produces behavior patterns similar to those attributed to mother–infant separation of both human and nonhuman primates. The additional finding from the first study, that multiple infant–infant separations appeared to stifle social maturation, should be viewed with caution in light of the absence of adequate control *S*s reared as a group of four

with no separations. However, a number of infant monkeys have been reared as infant–infant groups at the Wisconsin Laboratories, and data from these studies indicate that the maturational arrest found in the present experimentation, particularly in the social behaviors of cling and play, was unlikely to have resulted from the together-together housing alone. While these studies (e.g., Chamove, 1966; Harlow, 1961; Pratt, 1969) have generally found that clinging persists longer and play appears later and in lesser amounts in infant–infant groups as opposed to mother–infant or surrogate–infant groups, they did report drops in levels of cling and increases in the amount of play after the fourth month. Typical data, which were gathered from observations by Chamove (unpublished data, 1969) of two together-together raised groups of monkeys, are illustrated in Figure 8. These data stand in sharp contrast to the high level of cling and the virtual nonexistence of play at 9 mo. of the *S*s undergoing repetitive infant–infant separations (Figure 5). We believe that these behaviors, along with the decreases in locomotion and environmental exploration as well as the increases in self-mouthing, passivity, and self-clasping, indicate that the multiple separations had a profound stifling effect upon the social maturation of these *S*s.

The primary rationale given for the original mother–infant separation studies using nonhuman primates was to produce an experimental analog to the observations of Spitz and Bowlby of human infants separated from their mothers. Just as Spitz and Bowlby relied upon behavioral observations rather than inferred emotion or motivational states for their hypotheses and conclusions, the mother–infant monkey studies used direct observation of behavior as their sole criterion for drawing analogies. The same may be said for the present infant–infant studies. Spitz has often been criticized (e.g., Rie, 1966) for terming the human infant's reaction to maternal separation "anaclitic depression" by those who feel an infant lacks a sufficiently developed psyche to experience the depths of feeling attributed to adult human depression. Likewise, those engaging in research with subhuman *S*s who have demonstrated behavior analogous to anaclitic depression or neuroses in humans (e.g., Liddell, 1947; Masserman, 1943) have been soundly criticized for the terminology they employ to describe the behavioral reactions of their *S*s to a given experimental manipulation (e.g., Kubie, 1953). We feel that the issue involves not the behaviors observed but the analogies drawn from such studies. The behaviors occurred, and the analogies are only analogies. The same may be said concerning the relationship between infant–infant and mother–infant separation in monkeys. Although similar behavioral changes resulted from the two types of separations, we do not wish to equate an infant–infant bond with the mother–infant bond per se. A more useful strategy would be to consider both sets of studies as demonstrations of reaction to separation from an object of attachment rather than anything more specific.

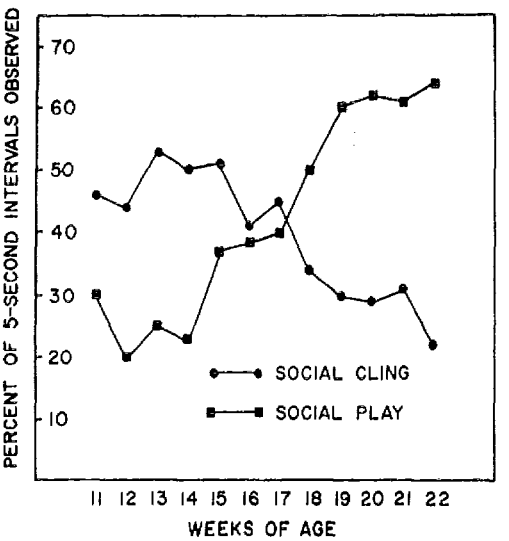


FIG. 8. Maturational relationship between social cling and play behavior for two groups of infant–infant reared monkeys not subjected to infant separation.

The technique of repetitive infant–infant separation was proposed as an alternative to the mother–infant model to study the effects of separation from an attachment object. In general, effects similar to those found in mother–infant separation studies were pro-

duced by each of the repetitive separations. In addition, it was found that *S*s did not adapt to the separation procedure and, if it were continued, failed to mature in their social behaviors. We feel that these studies illustrate the point that the mother–infant dyad need not provide the sole model for study of separation from an attachment object. Many of the advantages inherent in a mother–infant model are shared by other approaches, while several of the methodological disadvantages are not. Nor need this conclusion be limited to studies only of separation.

REFERENCES

BLOMQUIST, A. J., & HARLOW, H. F. The infant rhesus monkey program at the University of Wisconsin Primate Laboratory. *Proceedings of the Animal Care Panel,* 1961, 11, 57–64.

BOWLBY, J. The nature of a child's tie to his mother. *International Journal of Psychoanalysis,* 1958, 39, 1–24.

BOWLBY, J. Grief and mourning in infancy and early childhood. *Psychoanalytic Study of the Child,* 1960, 15, 9–52.

CHAMOVE, A. S. The effects of varying infant peer experience on social behavior in the rhesus monkey. Unpublished master's thesis, University of Wisconsin, 1966.

GEWIRTZ, J. L. A learning analysis of the effects of normal stimulation, privation, and deprivation on the acquisition of social motivation and attachment. In B. M. Foss (Ed.), *Determinants of infant behaviour.* London: Methuen, 1961.

HARLOW, H. F. The development of affectional patterns in infant monkeys. In B. M. Foss (Ed.), *Determinants of infant behaviour.* London: Methuen, 1961.

HARLOW, H. F., & HARLOW, M. K. The affectional systems. In A. M. Schrier, H. F. Harlow, & F. Stollnitz (Eds.), *Behavior of nonhuman primates.* Vol. 2. New York: Academic Press, 1965.

HINDE, R. A., SPENCER-BOOTH, Y., & BRUCE, M. Effects of 6-day maternal deprivation on rhesus monkey infants. *Nature,* 1966, 210, 1021–1033.

JENSEN, G. D. Reaction of monkey mothers to long-term separation from their infants. *Psychonomic Science,* 1968, 11, 171–172.

JENSEN, G. D., & TOLEMAN, C. W. Mother-infant relationship in the monkey, *Macaca nemestrina*: The effect of brief separation and mother-infant specificity. *Journal of Comparative and Physiological Psychology,* 1962, 55, 131–136.

JONES, E. E., & GERARD, H. B. *Foundations of social psychology.* New York: Wiley, 1967.

KAUFMAN, I. C., & ROSENBLUM, L. A. The reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. *Psychosomatic Medicine,* 1967, 29, 648–675.

KUBIE, L. S. The concept of normality and neurosis. In M. Heiman (Ed.), *Psychoanalysis and social work.* New York: International Universities Press, 1953.

LIDDELL, H. S. The experimental neurosis. *Annual Review of Physiology,* 1947, 9, 569–580.

MASSERMAN, J. H. *Behavior and neurosis.* Chicago: University of Chicago Press, 1943.

PRATT, C. L. The developmental consequences of variations in early social stimulation. Unpublished doctoral dissertation, University of Wisconsin, 1969.

RIE, H. E. Depression in childhood. *Journal of the American Academy of Child Psychiatry,* 1966, 5, 653–685.

SEAY, B., HANSEN, E., & HARLOW, H. F. Mother-infant separation in monkeys. *Journal of Child Psychology and Psychiatry,* 1962, 3, 123–132.

SEAY, B., & HARLOW, H. F. Maternal separation in the rhesus monkey. *Journal of Nervous and Mental Diseases,* 1965, 140, 434–441.

SHAFFER, H. R., & EMERSON, P. E. Development of social attachments in infancy. *Monographs of the Society for Research in Child Development,* 1964, 29, 1–77.

SPITZ, R. A. Anaclitic depression. *The Psychoanalytic Study of the Child,* 1946, 2, 313–347.

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